

Melih Dilbaz

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Education

Sabanci University | Bachelor of Engineering, Major in Computer Science June 2026

- **GPA:** 3.72/4.00 **Honors/Awards:** Full Scholarship (ranked 525th among 2.5 million participants)
- **Course Highlights:** Operating Systems, Deep Learning, Computer Architecture, Cloud Computing

Fellowship Programs:

Inzva hackerspace Fall 2024

- Ranked among the top 20 at season fall country-wide qualification
- Turkey's biggest competitive programming community

Akbank Fellowship Program 2022-2023

- Top 20 out of more than 10 thousands competitors
- Entrepreneurship | Sustainability | Mentourship | Inclusiveness

Work Experience

Lab Researcher – Software Engineering Intern, KU Leuven July-September 2025

- **Project:** [MV4QC](#) | Pipeline design for an industrial computer vision application.
- **Front-end:** Streamlit (Python) Web Application Tool
- **Back-end:** FastAPI – Python use – PostgreSQL DB

Data Science Intern, Vispera Information Technologies June-July 2025

Computer vision pipeline design for automated line detection. Assessed over 1000 real-case images and applied data augmentation strategies.

Technology Architecture Intern, Akbank Teknoloji August-September 2024

Gained experience in Data Science and Machine Learning, explored Chatbot Systems in finance.

Teaching Assistant, Sabanci University

- **Course:** Data Science (DSA210) 2025-2026
 - **Course:** Introduction to Probability (MATH203) Spring 2024
 - **Course:** Computational Approaches to Problem Solving (IF100) Fall 2023
- Assisted, and instructed 100 students with problem-solving techniques and supported course instruction.

Projects

Minecraft Server Management Platform on AWS Spring 2026

- Built a cloud-native Minecraft server management platform on AWS, integrating 14 AWS services, deploying a stateful game server on EC2 with encrypted EBS storage and S3-based backup/restore workflows.
- Designed the architecture, implemented IAM hardening, provisioned and configured the EC2 environment, system-based server setup, integrated EventBridge schedules for automated daily backups and monitoring

FORGE: On-Premise LLM Evaluation Framework 2025-2026

- Built a FastAPI/Next.js framework for LLM serving, benchmarking, and comparison.
- Integrated vLLM, HF-compatible model registry, and benchmarks such as MMLU, ARC-Easy, and BFCL Simple
- Added function-calling diagnostics and GPU metrics tracking for accuracy, throughput, VRAM usage, and deployment-cost analysis.

BlueSky Starter Packs Social Network – Echo Chamber Analysis (Data Science Analysis) Spring 2025

- Data: Populer Starter Packs from BlueSky API | Modularity by Leiden | InfoMap Technique
- Natural Science and Political Science starter packs and their influencers are cross-analyzed

Spending of a Student wrt. Inflation (Data Science Analysis) Fall 2024

- Data: Personal bank history - Official Inflation rates | ML modelling | Exploratory Data Analysis
- Analysis on the extent of the change in spending habits from that of discretionary to essential

HowUDoing - Messaging Application (Full-Stack mobile app. dev.) Fall 2024

- **Front-end:** React Native (HTML, CSS) (in progress)
- **Back-end:** SpringBoot – MongoDB (Authentication by JWT utilities)

Skills

- **Languages:** Turkish (Native), English (IELTS: 7.5 - 2025)
- **Technicals:** Python, C++, Java, SQL, JavaScript, React Native (HTML/CSS), PyTorch, Pandas, Android SDK (Mobile Application Development), MongoDB, SpringBoot, PostgreSQL

Activities & Volunteer Work

- **Work and Travel:** America (NJ) : amusement park – ride operator 2023-Summer (3 months)
- **CIP101 (2022 – Spring):** SU Course : Assisting in Primary Schools & visits to Special Education Schools